

UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA (UPSA)

DEGREE PROGRAMME

REGULAR SCHOOL



END OF SECOND SEMESTER EXAMINATIONS- 2018/2019 ACADEMIC YEAR

FACULTY OF INFORMATION TECHNOLOGY AND COMMUNICATION STUDIES

DEPARTMENT OF IT STUDIES

LEVEL 400

BITM 408: SOFTWARE QUALITY MANAGEMENT

Date: Wednesday, May 15, 2019

TIME ALLOWED: 2 HOURS

INDEX NUMBER

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INSTRUCTION(S):

ATTEMPT ALL QUESTIONS

THERE ARE TWO (2) SECTIONS IN THIS PAPER, SECTION A AND B.

SECTION A IS COMPULSORY

**ANSWER QUESTION 1 AND ANY OTHER QUESTION FROM SECTION B
BOTH SECTIONS MUST BE ANSWERED IN THE BOOKLET PROVIDED.**

SECTION A

Instructions: Answer all questions in this section [1 Mark each]

For Questions 1 to 5, complete the sentences or statements provided with the appropriate words

1. The termination of the ability of a software to perform a required function or its inability to perform within previously specified limits is known as a software.....
2. The degree to which a software module or other work product can be used in more than one computing program or software system is known as.....
3. Cost of warranty execution and damage caused to a corporate image as well as the cost of losing clients is known as.....
4. Computer programs, procedures, and possibly associated documentation and data pertaining to the operation of a computer system is known as.....
5.is the degree to which a system or component is free from faults in its specification, design, and implementation.

Determine whether the following statements are TRUE or FALSE

6. ISO 13485 is a specific standard for Medical Devices
7. The degree of effectiveness and efficiency with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts, or to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified is known as Modularity.
8. Non-compliance with current processes/procedures poses no problem as far as software error is concerned
9. A Quality Model is a defined set of characteristics, and of relationships between them, which provides a framework for specifying quality requirements and evaluating quality.
10. Good team spirit can foster software quality

SECTION B

Instructions: Answer Question 1 [30 marks] and any other question [20 marks]

Question 1 [30 Marks]

- a. What
 - i. is Agile testing and what is the importance of Agile testing? **(5 Marks)**
 - ii. are the five common solutions for software developments problems? **(10 Marks)**
 - iii. is the difference between UAT (User Acceptance Testing) and System testing? **(4 Marks)**
- b. Explain the steps for Bug Cycle? **(7 Marks)**
- c. Briefly explain any 4 roles of Software Quality Assurance engineer? **(2 Marks)**
- d. Outline any four roles of the moderator in a software inspection process **(2 Marks)**

Question 2 [20 Marks]

- a. The software requirement document for the tender for development of “Super-lab,” a software system for managing a hospital laboratory, consists of chapters according to the required quality factors as follows: correctness, reliability, efficiency, integrity, usability, maintainability, flexibility, testability, portability, reusability and interoperability. In the following table you will find sections taken from the mentioned requirements document. For each section, fill in the name of the factor that best fits the requirement (chose only one factor per requirements section).

No	Section taken from the software requirement document	The requirements factor
1	The probability that the “Super-lab” system software will fail during normal operating hours (9 a.m. to 4 p.m.) must be less than 0.5%.	
2	The “Super-lab” software system will enable direct transfer of laboratory results to those files of hospitalized patients managed by the “MD-File” software package. Interoperability	
3	The “Super-lab” software system will include a module that prepares a detailed report of the patient’s laboratory test results during his current hospitalization. (This report will serve as an appendix to the family physician’s file.) The time required to obtain this printed report will be less than 60 seconds; the level of accuracy and completeness will be at least 99%.	
4	The “Super-lab” software to be developed for hospital laboratory use may be adapted later for private laboratory use.	

5	The training of a laboratory technician, requiring no more than 3 days, will enable the technician to reach level C of “Super-lab” software usage. This means that he or she will be able to manage reception of 20 patients per hour.	
6	The “Super-lab” software system will record a detailed users’ log. In addition, the system will report attempts by unauthorized persons to obtain medical information from the laboratory test results database. The report will include the following information: the network identification of the applying terminal, the system code of the employee who requested that information, the day and time of attempt and the type of attempt.	
7	The “Super-lab” subsystem that deals with billing patients for their tests may be eventually used as a subsystem in the “Physiotherapy Center” software package.	
8	The “Super-lab” software system will process all the monthly reports for the hospital departments’ management, the hospital management, and the hospital controller according to the development contract.	
9	The software system should be able to serve 12 workstations and 8 automatic testing machines with a single model AS20 server and a CS25 communication server that will be able to serve 25 communication lines. This hardware system should conform to all availability requirements.	
10	The “Super-lab” software package developed for the Linux operating system should be compatible for applications in a Windows NT environment.	

[10 Marks]

- b. Differentiate between white box and black box testing **(4 marks)**
- c. State and explain the differences between the two major categories of verification testing **(4 marks)**

Question 3 [20 Marks]

- a. Briefly differentiate between software quality, software quality control, and software quality assurance **[6 Marks]**
- b. Briefly differentiate between software errors, software defects, and software failure **[6 Marks]**
- c. A requirement is said to be of good quality when it meets a number of characteristics. List and briefly explain 8 of them **[8 Marks]**